

Sheaves & Proofs

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Desired background

Some exposure to proof theory (sequent calculus)
and semantics (structures, satisfaction, soundness, completeness)
for classical predicate logic.

Not required background

- Intuitionistic logic (sequent calculus, Kripke models)
- Basic category theory (categories, functors, natural transformations)
- Basic point-set topology (topologies, covers)

Part 1

Coverage Semantics over Partial Orders

(The logic of localic toposes)

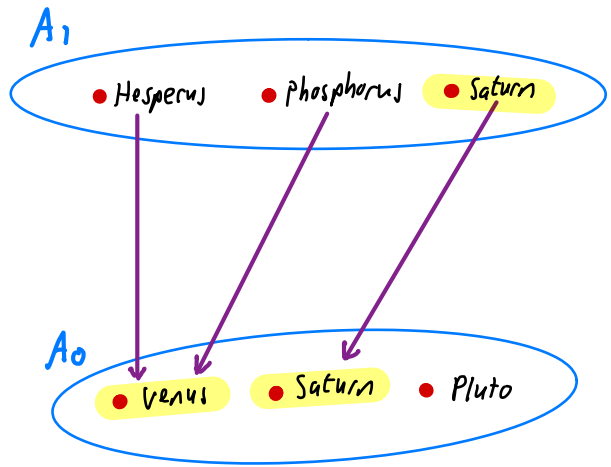
- Kripke semantics
- Coverage semantics
- Cut-elimination via
Constructive completeness
- The **77**-dense coverage.

(Inverted) Kripke semantics

A Kripke Model is given by

- A partial order $(W, \leq)$ of Worlds
- To every $w \in W$ a set A_w
- Transition functions $(t_{vw} : A_w \rightarrow A_v)_{v \leq w}$
Satisfying $t_{ww} = id_{A_w}$; $u \leq v \leq w \Rightarrow t_{uw} = t_{uv} \circ t_{vw}$
- For every predicate P of arity k
Subsets $P_w \subseteq (A_w)^k$ satisfying
 $v \leq w \Rightarrow t_{vw}(P_w) \subseteq P_v$.

$$W = \begin{array}{c} 1 \\ | \\ 0 \end{array}$$



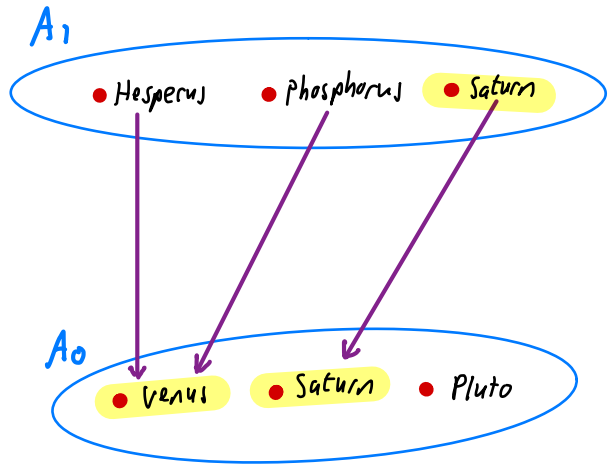
 The planet predicate

Category-theoretic Formulation

A Kripke model is given by

- A partial order $(W, \leq)$ of worlds
- A functor $A: W^{op} \rightarrow \underline{\text{Set}}$
(i.e., a presheaf $A \in \text{Psh}(W)$)
- For every predicate P of arity k a subpresheaf $P \subseteq A^k$

$$W = \begin{array}{c} \bullet^1 \\ | \\ \bullet^0 \end{array}$$



 The planet predicate

Notation

$\text{Venus} = t_{01}(\text{Hesperus})$	(previous slide)
$= A(0<1)(\text{Hesperus})$	(this slide)
$= \text{Hesperus} \cdot 0$	(henceforth)

Kripke forcing relation

$$w \Vdash \phi$$

$$w \Vdash P(a_1, \dots, a_k) \Leftrightarrow (a_1, \dots, a_k) \in P_w$$

$$w \Vdash a_1 = a_2 \Leftrightarrow a_1 = a_2$$

$$w \Vdash \phi \wedge \psi \Leftrightarrow w \Vdash \phi \text{ and } w \Vdash \psi$$

$$w \Vdash \phi \vee \psi \Leftrightarrow w \Vdash \phi \text{ or } w \Vdash \psi$$

$$w \Vdash \perp \Leftrightarrow \text{false}$$

$$w \Vdash \exists x \phi \Leftrightarrow \text{there exists } a \in A_w \text{ s.t. } w \Vdash \phi[x:=a]$$

$$w \Vdash \phi \rightarrow \psi \Leftrightarrow \text{for all } v \leq w \quad v \Vdash \phi \cdot v \text{ implies } v \Vdash \psi \cdot v$$

$$w \Vdash \forall x \phi \Leftrightarrow \text{for all } v \leq w \text{ and } a \in A_v \quad v \Vdash (\phi \cdot v)[x:=a]$$

$1 \Vdash \text{Hesperus} = \text{Phosphorus}$

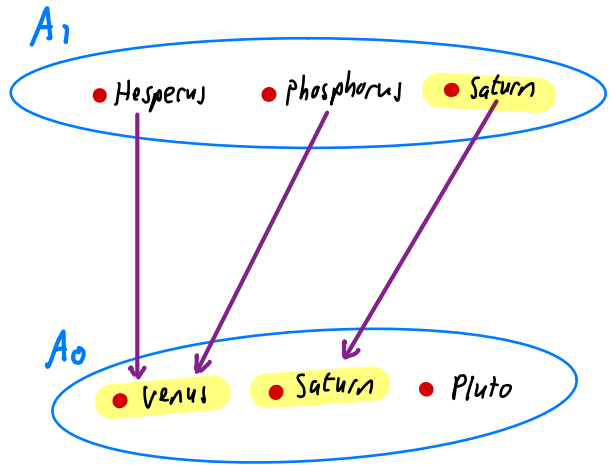
$0 \Vdash \text{Hesperus} \cdot 0 = \text{Phosphorus} \cdot 0$

$1 \Vdash \neg (\text{Hesperus} = \text{Phosphorus})$

$1 \Vdash H = P \vee \neg (H = P)$

$1 \Vdash \neg \neg (H = P)$

$$W = \begin{matrix} & 1 \\ & \vdots \\ & 0 \end{matrix}$$



 The planet predicate

$\neg \phi := \phi \rightarrow \perp$

$w \Vdash \neg \phi \Leftrightarrow \forall v \leq w. v \nVdash \phi$

$w \Vdash \neg \neg \phi \Leftrightarrow \forall v \leq w. \neg \neg \exists u \leq v. u \Vdash \phi$

Meta theorems for Kripke semantics

Monotonicity If $w \Vdash \phi$ and $v \leq w$ then $v \Vdash \phi$.

Soundness If ϕ is provable in intuitionistic predicate logic then for all Kripke models $(W, \dots)$ and $w \in W$, $w \Vdash \phi$.

Completeness If, for all Kripke models $(W, \dots)$ and $w \in W$, $w \Vdash \phi$ then ϕ is provable in intuitionistic predicate logic.

Completeness is not provable in a constructive meta-theory.

To achieve constructive completeness, we need more general notion of model.

Coverage base

a.u.a. base for a
Grothendieck topology
on a poset

A coverage base on a poset $\mathbb{W}$ is a relation $\triangleright$ relating pairs $C \triangleright w$, where $w \in \mathbb{W}$ and $C \subseteq \downarrow w$, satisfying:

$$\downarrow w := \{v \in \mathbb{W} \mid v \leq w\}$$

$$\triangleright \subseteq \sum_{w \in \mathbb{W}} \mathcal{P}(\downarrow w)$$

Reflexivity $\{w\} \triangleright w$

Transitivity If $C \triangleright w$ and, for all $v \in C$, $C_v \triangleright v$ then $\bigcup_{v \in C} C_v \triangleright w$.

Stability If $C \triangleright w$ and $w' \leq w$ then there exists $D \triangleright w'$ such that, for all $v' \in D$ there exists $v \in C$ such that $v' \leq v$.

Example coverage bases

① The identity coverage $\{w\} \triangleright w$ on any poset W

② For any topological space T , define

$$W = \mathcal{O}(T)$$

(open sets)

$$v \leq w \iff v \subseteq w$$

$$C \triangleright w \iff \bigcup C = w$$

(C covers w)

Exercise Prove that ② satisfies properties of a cover base!

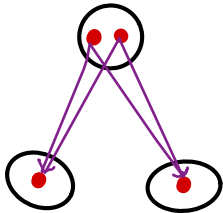
A separated Model is given by

- A partial order $(W, \leq)$ of worlds with coverage base $\triangleright$
- To every $w \in W$ a set A_w
- Transition functions $(t_{vw} : A_w \rightarrow A_v)_{v \leq w}$ s.t. $u \leq v \leq w \Rightarrow t_{uw} = t_{uv} \circ t_{vw}$
- $C \triangleright w \Rightarrow \forall a, b \in A_w (\forall v \in C, t_{vw}(a) = t_{vw}(b)) \Rightarrow a = b$. ($\triangleleft$ -separated)
- For every predicate P of arity k subsets $P_w \subseteq (A_w)^k$ satisfying
 - $\forall \vec{a} \in A_w^k (\vec{a} \in P_w \Rightarrow \forall v \leq w, t_{vw}(\vec{a}) \in P_v)$
 - $C \triangleright w \Rightarrow \forall \vec{a} \in A_w^k (\forall v \in C, t_{vw}(\vec{a}) \in P_v) \Rightarrow \vec{a} \in P_w$ (P is $\triangleleft$ -closed)

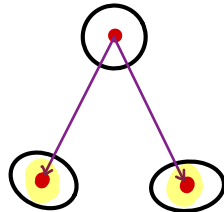
Category-theoretic Formulation

A separated model is given by

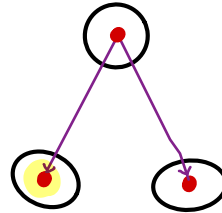
- A partial order $(W, \leq)$ of worlds
- A $\triangleright$ -separated presheaf $A : W^{op} \rightarrow \underline{\text{Set}}$
- For every predicate P of arity k , a $\triangleright$ -closed subpresheaf $P \subseteq A^k$



not separated



not closed



a separated model

$$W = \bigwedge_{0 \ 0'}^1$$

$$\{0, 0'\} \triangleright 1$$

$$\{0\} \not\triangleright 1$$

$$\{0'\} \not\triangleright 1$$

New forcing clauses

$w \Vdash \phi \vee \psi \iff$ there exists a cover $C \triangleright w$ such that,
for every $v \in C$, $v \Vdash \phi \cdot v$ or $v \Vdash \psi \cdot v$

$w \Vdash \perp \iff \emptyset \triangleright w$

$w \Vdash \exists x \phi \iff$ there exists a cover $C \triangleright w$ such that,
for every $v \in C$,
there exists $a \in \underline{A}_v$ s.t. $v \Vdash (\phi \cdot v)[x := a]$

N.B.

- The clauses for predicates, $=$, $\wedge$, $\rightarrow$, $\forall$ are as for Kripke semantics

For our example coverage bases

(1) The identity coverage base $\{w\} \triangleright w$ on any poset W

Recovers Kripke semantics

(2) The topological cover relation for $W = \mathcal{O}(T)$.

Recovers topological semantics of intuitionistic logic

Meta theorems for coverage semantics

Monotonicity If $w \Vdash \phi$ and $v \leq w$ then $v \Vdash \phi$.

Cover property If $C \supseteq w$ satisfies, for all $v \in C$, $v \Vdash \phi$,
then $w \Vdash \phi$.] See warning at the end of Part 2

Soundness If ϕ is provable in intuitionistic predicate logic
then for all separated models $(W, \dots)$ and $w \in W$, $w \Vdash \phi$.

Completeness If, for all separated models $(W, \dots)$ and $w \in W$, $w \Vdash \phi$
then ϕ is provable in intuitionistic predicate logic

Completeness is now provable in a constructive meta-theory!

Intuitionistic sequent Calculus

Predicate symbols $P(x_1, \dots, x_n)$ ($n \geq 0$)

No function symbols (in particular no constants) (for brevity of exposition)

Equality $t_1 = t_2$

[The only terms are variables, but we use $t, t_1, \dots$ as meta-variables in places where arbitrary terms would be allowed in presence of function symbols.]

Sequent: $\Theta; \Gamma \Rightarrow \psi$

where Θ is finite set of variables

Γ is finite set of Θ -formulas, ψ is Θ -formula

A Θ -formula is a formula whose free variables are contained in Θ .

Rules

$$\frac{}{\Theta ; \Gamma, P(\vec{t}) \Rightarrow P(\vec{t})}$$

$$\frac{\Theta ; \Gamma, \phi, \phi' \Rightarrow \psi}{\Theta ; \Gamma \ni \phi \wedge \phi' \Rightarrow \psi}$$

$$\Theta ; \Gamma \ni \phi \wedge \phi' \Rightarrow \psi$$

$$\frac{\Theta ; \Gamma \Rightarrow \psi \quad \Theta ; \Gamma \Rightarrow \psi'}{\Theta ; \Gamma \Rightarrow \psi \wedge \psi'}$$

$$\Theta ; \Gamma \Rightarrow \psi \wedge \psi'$$

$$\frac{\Theta ; \Gamma, \phi \Rightarrow \psi \quad \Theta ; \Gamma, \phi' \Rightarrow \psi}{\Theta ; \Gamma \ni \phi \vee \phi' \Rightarrow \psi}$$

$$\Theta ; \Gamma \ni \phi \vee \phi' \Rightarrow \psi$$

$$\frac{\Theta ; \Gamma \Rightarrow \psi}{\Theta ; \Gamma \Rightarrow \psi \vee \psi'}$$

$$\Theta ; \Gamma \Rightarrow \psi \vee \psi'$$

$$\frac{\Theta ; \Gamma \Rightarrow \psi'}{\Theta ; \Gamma \Rightarrow \psi \vee \psi'}$$

$$\Theta ; \Gamma \Rightarrow \psi \vee \psi'$$

$$\frac{}{\Theta ; \Gamma \ni \perp \Rightarrow \psi}$$

$$\frac{\Theta ; \Gamma, \phi' \Rightarrow \psi \quad \Theta ; \Gamma \Rightarrow \phi}{\Theta ; \Gamma \ni \phi \rightarrow \phi' \Rightarrow \psi}$$

$$\Theta ; \Gamma \ni \phi \rightarrow \phi' \Rightarrow \psi$$

$$\frac{\Theta ; \Gamma, \psi \Rightarrow \psi'}{\Theta ; \Gamma \Rightarrow \psi \rightarrow \psi'}$$

$$\Theta ; \Gamma \Rightarrow \psi \rightarrow \psi'$$

$$\frac{\Theta \wp x; \Gamma, \phi \Rightarrow \psi}{\Theta; \Gamma \ni \exists x \phi \Rightarrow \psi}$$

$$\frac{\Theta; \Gamma' \Rightarrow \psi[t/x]}{\Theta; \Gamma \Rightarrow \exists x \psi}$$

$$\frac{\Theta; \Gamma', \phi[t/x] \Rightarrow \psi}{\Theta; \Gamma \ni \forall x t \Rightarrow \psi}$$

$$\frac{\Theta \wp x; \Gamma \Rightarrow \psi}{\Theta; \Gamma \Rightarrow \forall x \psi}$$

$$\frac{\Theta; \Gamma \Rightarrow \psi[s/z]}{\Theta; \Gamma \ni s=t \Rightarrow \psi[t/z]}$$

$$\frac{\Theta; \Gamma \Rightarrow \psi[t/z]}{\Theta; \Gamma \ni s=t \Rightarrow \psi[s/z]}$$

$$\frac{}{\Theta; \Gamma \Rightarrow t=t}$$

$$\frac{\Theta; \Gamma \Rightarrow \phi \quad \Theta; \Gamma, \phi \Rightarrow \psi}{\Theta; \Gamma \Rightarrow \psi} \text{ cut}$$

Define $\Theta; \Gamma \vdash \psi$ $:\Leftrightarrow$ the sequent $\Theta; \Gamma \Rightarrow \psi$ is provable

$\Theta; \Gamma \vdash_{cf} \psi$ $:\Leftrightarrow$ the sequent $\Theta; \Gamma \Rightarrow \psi$ is cut-free provable.

Lemma (admissible rules with/without cut)

- (identity) $\Theta; \Gamma, \psi \vdash_{cf} \psi$
- (weakening) $\Theta; \Gamma \vdash \psi \Rightarrow \Theta, \theta'; \Gamma, \Gamma' \vdash \psi$
 $\Theta; \Gamma \vdash_{cf} \psi \Rightarrow \Theta, \theta'; \Gamma, \Gamma' \vdash_{cf} \psi$

Proofs Straightforward inductions.

Theorem (Cut admissibility) $\Theta; \Gamma \vdash_{cf} \phi$ and $\Theta; \Gamma, \phi \vdash_{cf} \psi \Rightarrow \Theta; \Gamma \vdash_{cf} \psi$

We shall give a semantic proof, by establishing

Soundness $\Theta; \Gamma \vdash \psi \Rightarrow \Theta; \Gamma \models \psi$

Cut-free completeness $\Theta; \Gamma \models \psi \Rightarrow \Theta; \Gamma \vdash_{cf} \psi$

where $\models$ is semantic entailment w.r.t. separated models.

The proof will be carried out in a constructive meta-theory, so (in principle) it's possible to extract from the proof an algorithm that maps derivations with cuts to cut-free derivations.

Related work: Beth, Veldman et.al., Palmgren, Avigad

Semantic entailment

$$\Theta; \Gamma \models \psi$$

$:\Leftrightarrow$

for every poset W and coverage base $\triangleright$ on W ,

for every separated model $(A: W^{op} \rightarrow \underline{\text{set}}, \{p_w\}_w)$,

for every $w \in W$,

for every $\rho: \Theta \rightarrow A_w$

if, for every $\phi \in \Gamma$, $w \Vdash \phi[\rho]$,

then $w \Vdash \psi[\rho]$

$\forall w$

$\Theta; \Gamma \models_m \psi$

Proof of soundness: $\Theta; \Gamma \vdash \psi \Rightarrow \Theta; \Gamma \models \psi$

Consider any m , (i.e., $\mathbb{W}, \triangleright$ and separated model $A, \dots$)

We prove by ind. on derivation of $\Theta; \Gamma \vdash \psi$ that $\Theta; \Gamma \models_m \psi$ holds

Exercise!

Proof of cut-free completeness: $\Theta; \Gamma \models \psi \Rightarrow \Theta; \Gamma \vdash_{cf} \psi$

In Part 1, we consider just the case of intuitionistic propositional logic

- All predicate symbols have arity 0.
(Predicates of arity 0 are atomic propositions.)
- No variables, no quantifiers $\forall, \exists$
- No equality $=$

We build a specific separated model $\mathcal{M}_{ctx}$

The modal M_{ctx} (propositional logic)

$W_{ctx} := \{ \Gamma \mid \Gamma \text{ a finite set of formulas} \}$ (The set of contexts)

$\Gamma' \leq \Gamma : \Leftrightarrow \Gamma' \supseteq \Gamma$

$C \triangleright_{ctx} \Gamma : \Leftrightarrow \frac{C}{\Gamma} \text{ is a derivable context rule}$

$A_{\Gamma} := \emptyset$

$() \in P_{\Gamma} : \Leftrightarrow \Gamma \vdash_{cf} P \quad . \quad (\Gamma \Vdash P : \Leftrightarrow \Gamma \vdash_{cf} P)$

○ Corrected from lecture. (Thanks to Dominik Wehr)

Context rules

$$\frac{\Gamma, \phi, \phi'}{\Gamma \ni \phi \wedge \phi'}$$

$$\frac{\Gamma, \phi \quad \Gamma, \phi'}{\Gamma \ni \phi \vee \phi'}$$

$$\frac{}{\Gamma \ni \perp}$$

$$\frac{\Gamma, \phi'}{\Gamma \ni \phi \rightarrow \phi'} \quad \Gamma \vdash_{cf} \phi$$

For any context Γ and set of contexts C ,

$\frac{C}{\Gamma}$ is a derivable context rule if there exists a derivation tree of contexts with conclusion Γ , in which every leaf is in C .

Lemma Suppose $\frac{C}{\Gamma}$ is a derivable context rule.

- (weakening) For any Γ'' ,

$$\frac{\{\Gamma', \Gamma'' \mid \Gamma' \in C\}}{\Gamma, \Gamma''}$$

is a derivable context rule

The stability
of $\triangleright_{\text{ctx}}$
follows.

- (sequencing) For any ψ ,

$$\frac{\{\Gamma' \Rightarrow \psi \mid \Gamma' \in C\}}{\Gamma \Rightarrow \psi}$$

is a cut-free derivable sequent rule.

The Δ -closed
property of
 M_{ctx}
follows

Main lemma $\mathcal{M}_{ctx}$ enjoys the following.

For any $\Gamma \in W_{ctx}$ and formula ϕ

(1) if $\phi \in \Gamma$ then $\Gamma \Vdash \phi$.

(2) if $\Gamma \Vdash \phi$ then $\Gamma \vdash_{cf} \phi$.

Proof of cut-free completeness: $\Gamma \models \psi \Rightarrow \Gamma \vdash_{cf} \psi$

Suppose $\Gamma \models \psi$.

In particular $\Gamma \models_{\mathcal{M}_{ctx}} \psi$. $\otimes$

By ① of main lemma, for every $\phi \in \Gamma$, $\Gamma \Vdash \phi$.

By $\otimes$, $\Gamma \Vdash \psi$.

By ② of main lemma, $\Gamma \vdash_{cf} \psi$.

□

Main lemma M_{ctx} enjoys the following.

For any $\Gamma \in W_{ctx}$ and formula ϕ

(1) if $\phi \in \Gamma$ then $\Gamma \Vdash \phi$.

(2) if $\Gamma \Vdash \phi$ then $\Gamma \vdash_{cf} \phi$.

Proof

(1) and (2) are proved simultaneously by induction on ϕ .

□

Case $\phi \equiv \phi_1 \rightarrow \phi_2$

(1) $\phi_1 \rightarrow \phi_2 \in \Gamma \Rightarrow \Gamma \Vdash \phi_1 \rightarrow \phi_2$

Suppose $\Gamma' \supseteq \Gamma$ and $\Gamma' \Vdash \phi_1$,

By i.h. (2) $\Gamma' \vdash_{cf} \phi_1$

By $\bigcirc$, $\{\Gamma', \phi_2\} \supseteq_{ctx} \Gamma'$

By i.h. (1), $\Gamma', \phi_2 \Vdash \phi_2$

By closing property $\Gamma' \Vdash \phi_2$.

Case $\phi \equiv \phi_1 \rightarrow \phi_2$

$$(2) \quad \Gamma \Vdash \phi_1 \rightarrow \phi_2 \Rightarrow \Gamma \vdash_{cf} \phi_1 \rightarrow \phi_2$$

By i.h. (1) $\Gamma, \phi_1 \Vdash \phi_1$

By $\circ$, $\Gamma, \phi_1 \Vdash \phi_2$

By i.h. (2) $\Gamma, \phi_1 \vdash_{cf} \phi_2$

Hence $\Gamma \vdash_{cf} \phi_1 \rightarrow \phi_2$ by $(\rightarrow R)$.

Case $\phi \equiv \phi_1 \vee \phi_2$

(1) $\phi_1 \vee \phi_2 \in \Gamma \Rightarrow \Gamma \Vdash \phi_1 \vee \phi_2$

We have $\{(\Gamma, \phi_1), (\Gamma, \phi_2)\} \triangleright_{ctx} \Gamma$

By i.h. (1), $\Gamma, \phi_1 \Vdash \phi_1$ and $\Gamma, \phi_2 \Vdash \phi_2$.

Hence $\Gamma \Vdash \phi_1 \vee \phi_2$.

Case $\phi \equiv \phi_1 \vee \phi_2$

$$(2) \quad \Gamma \Vdash \phi_1 \vee \phi_2 \Rightarrow \Gamma \vdash_{cf} \phi_1 \vee \phi_2$$

By $\circ$, there is $C \triangleright_{ctx} \Gamma$ such that

for every $\Gamma' \in C$, $\Gamma' \Vdash \phi_1$ or $\Gamma' \Vdash \phi_2$.

So for every $\Gamma' \in C$, $\Gamma' \vdash_{cf} \phi_1$ or $\Gamma' \vdash_{cf} \phi_2$, by i.h (2).

So for every $\Gamma' \in C$, $\Gamma' \vdash_{cf} \phi_1 \vee \phi_2$, by ($\vee R$) rules.

Since $\frac{C}{\Gamma}$ is admissible, $\Gamma \vdash_{cf} \phi_1 \vee \phi_2$, by sequencing property.



Sieve

In a coverage (rather than coverage base) we only allow covers $C \triangleright w$, where C is a sieve on w .

A set $C \subseteq W$ is a sieve on $w \in W$ if

- $C \subseteq \downarrow w$
- $C = \downarrow C$ (C is down closed)

If C is a sieve on w and $v \leq w$ then

$$C \cdot v := \{u \in C \mid u \leq v\}$$

is a sieve on v .

$$w \mapsto \text{Sieves}(w)$$

defines a presheaf

$$W^{op} \rightarrow \underline{\text{Set}}$$

Coverage

A coverage is a relation $C \triangleright w$ between $w \in W$ and sieves C on w satisfying

Reflexivity $\downarrow w \triangleright w$

Transitivity If $C \triangleright w$ and S is a sieve on w satisfying

$$\forall v \in C, S \cdot v \triangleright v$$

then $S \triangleright w$.

Stability If $C \triangleright w$ and $v \leq w$ then $C \cdot v \triangleright v$.

Proposition (Properties of coverage)

Let $\triangleright$ be a coverage. Then $\triangleright$ satisfies

Monotonicity If $C \triangleright w$ and S is a sieve on w then

$$C \subseteq S \Rightarrow S \triangleright w$$

Intersection If $C \triangleright w$ and $C' \triangleright w$ then $C \cap C' \triangleright w$

Proof hints For monotonicity use reflexivity + transitivity.

For intersection use transitivity + stability.

Coverages vs. Coverage bases

Proposition

(1) If $\triangleright$ is a coverage base then

$$C \triangleright w \quad :\Leftrightarrow \quad \exists B \subseteq C, B \triangleright w \text{ and } C = \downarrow B$$

is a coverage. We say $\triangleright$ is the coverage generated by $\triangleright$.

(2) Every coverage $\triangleright$ is generated by the coverage base

$$B \triangleright w \quad :\Leftrightarrow \quad \downarrow B \triangleright w.$$

Proof Exercise!

The $\neg\neg$ -dense coverage

On any poset W define

$$C \triangleright_{\neg\neg} w \iff C \text{ is a sieve on } w \text{ and} \\ \forall v \leq w \neg\neg \exists u \leq v \ u \in C.$$

Proposition $\triangleright_{\neg\neg}$ is a coverage.

Proof Exercise.

$\neg\neg \text{Dense} \Rightarrow \text{Classical}$

Proposition Using the coverage $\triangleright_{\neg\neg}$, $\forall w \in W$ with $\neg\neg\phi \rightarrow \phi$

Proof Suppose $w \Vdash \neg\neg\phi$

we show $\{u \leq w \mid u \Vdash \phi\} \triangleright_{\neg\neg} w$. Then $w \Vdash \phi$ by cover property.

$\{u \leq w \mid u \Vdash \phi\}$ is a sieve by monotonicity of $\Vdash$

Suppose $v \leq w$. We need to show $\neg\neg \exists u \leq v, u \Vdash \phi$.

Suppose for $\perp$ that $\neg \exists u \leq v, u \Vdash \phi$.

Then $v \Vdash \neg\phi$.

By $\circ$, $v \Vdash \perp$. So $\emptyset \triangleright_{\neg\neg} v$. So $\forall v' \leq v$ $\neg\neg \exists u' \leq v, u' \Vdash \phi$. I.e. $\perp$.



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Part 2

Sheaf Semantics over Categories

(The logic of Grothendieck toposes)

- Presheaf semantics
- (coverage based) sheaf semantics
- Dense coverages
- Atomic coverages

Let us attempt to directly generalise our Cut-free Completeness proof from propositional logic to intuitionistic predicate logic

We try to define appropriate W_{ctx} and $\triangleright_{\text{ctx}}$

$W_{\text{ctx}} := \{ \Theta; \Gamma \mid \Theta \text{ a finite set of variables, } \Gamma \text{ a finite set of } \Theta\text{-formulas} \}$

The set of contexts

$$\Theta'; \Gamma' \leq \Theta; \Gamma \iff \Theta'; \Gamma' \supseteq \Theta; \Gamma$$

$$C \triangleright_{\text{ctx}} \Theta; \Gamma \iff \frac{C}{\Theta; \Gamma} \text{ is a derivable context rule}$$

Context rules

$$\frac{\Theta; \Gamma, \phi, \phi'}{\Theta; \Gamma \ni \phi \wedge \phi'}$$

$$\frac{\Theta \oplus x; \Gamma, \phi}{\Theta; \Gamma \ni \exists x \phi}$$

$$\frac{\Theta; \Gamma', \phi \quad \Theta; \Gamma, \phi'}{\Theta; \Gamma \ni \phi \vee \phi'}$$

$$\frac{\Theta; \Gamma', \phi[t/x]}{\Theta; \Gamma \ni \forall x \phi}$$

$$\frac{}{\Theta; \Gamma \ni \perp}$$

$$\frac{\Theta; \Gamma, \phi'}{\Theta; \Gamma \ni \phi \rightarrow \phi'} \quad \Theta; \Gamma \vdash_{cf} \phi$$

However, the weakening property fails for derivable context rules.

E.g. $\frac{y ; \exists x P(x), P(y)}{; \exists x P(x)}$ is a derivable context rule

but $\frac{y ; \exists x P(x), P(y), Q(y)}{y ; \exists x P(x), Q(y)}$ is not a derivable context rule

Indeed $\triangleright_{\text{ctx}}$ does not satisfy the stability property of a coverage base.

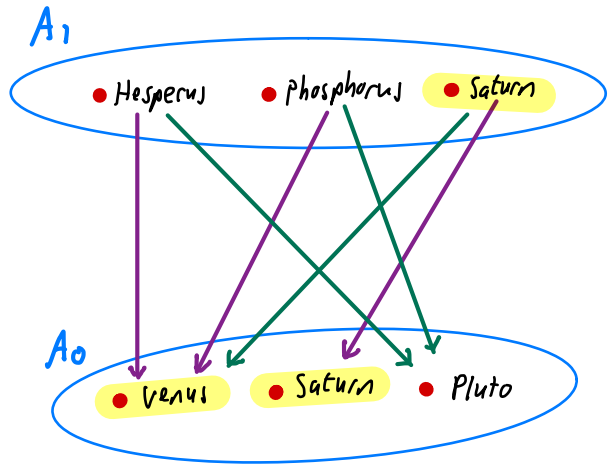
An elegant solution to this problem is to replace the partial order $\mathbb{W}_{\text{ctx}}$ with a category.

Presheaf semantics

A Presheaf Model is given by

- A (small) category $\mathbb{C}$ of Worlds
- To every $X \in \mathbb{C}$ a set A_X
- Transition functions $(t_f : A_X \rightarrow A_Y)_{Y \xrightarrow{f} X}$
Satisfying $Z \xrightarrow{g} Y \xrightarrow{f} X \Rightarrow t_{f \circ g} = t_g \circ t_f$
- For every predicate P of arity k
Subsets $P_w \subseteq (A_w)^k$ satisfying
 $Y \xrightarrow{f} X \Rightarrow t_f(P_X) \subseteq P_Y$

$$\mathbb{C} = \begin{matrix} & & 1 \\ & \nearrow f_0 & \nearrow f_1 \\ 0 & & \end{matrix}$$



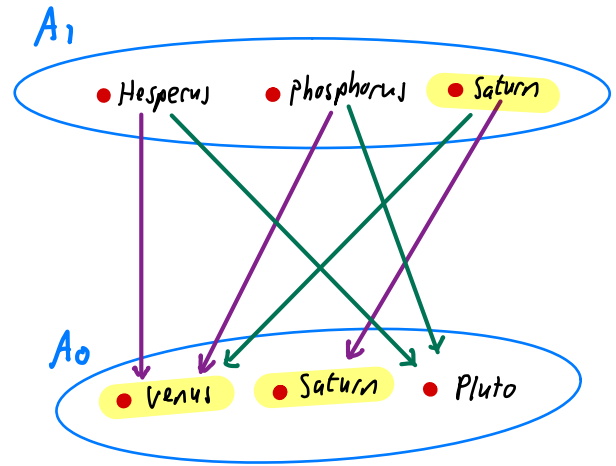
 The planet predicate

Category-theoretic Formulation

A Presheaf model is given by

- A small category $\mathcal{C}$ of worlds
- A functor $A: \mathcal{C}^{op} \rightarrow \underline{\text{Set}}$
(i.e., a presheaf $A \in \text{Psh}(\mathcal{C})$)
- For every predicate P of arity k
a subpresheaf $R \subseteq A^k$

$$\mathcal{C} = \begin{matrix} & & 1 \\ & \nearrow f_0 & \searrow f_1 \\ 0 & & 0 \end{matrix}$$



 The planet predicate

Notation

$\text{Venus} = t_{f_0}(\text{Hesperus})$ (previous slide)
 $= A(f_0)(\text{Hesperus})$ (this slide)
 $= \text{Hesperus} \cdot f_0$ (henceforth)

Presheaf forcing relation

$$X \Vdash \phi$$

$$X \Vdash P(a_1, \dots, a_k) \iff (a_1, \dots, a_k) \in P_x$$

$$X \Vdash a_1 = a_2 \iff a_1 = a_2$$

$$X \Vdash \phi \wedge \psi \iff X \Vdash \phi \text{ and } X \Vdash \psi$$

$$X \Vdash \phi \vee \psi \iff X \Vdash \phi \text{ or } X \Vdash \psi$$

$$X \Vdash \perp \iff \text{false}$$

$$X \Vdash \exists x \phi \iff \text{there exists } a \in A_x \text{ s.t. } X \Vdash \phi[x := a]$$

$$X \Vdash \phi \rightarrow \psi \iff \text{for all } Y \xrightarrow{f} X \quad Y \Vdash \phi \cdot f \text{ implies } Y \Vdash \psi \cdot f$$

$$X \Vdash \forall x \phi \iff \text{for all } Y \xrightarrow{f} X \text{ and } a \in A_Y \quad Y \Vdash (\phi \cdot f)[x := a]$$

Meta theorems for Presheaf Semantics

Monotonicity If $x \Vdash \phi$ and $y \xrightarrow{f} x$ then $y \Vdash \phi$.

Soundness If ϕ is provable in intuitionistic predicate logic
then for all **presheaf models** $(\mathbb{C}, \dots)$ and $x \in \mathbb{C}$, $x \Vdash \phi$.

Completeness If, for all **presheaf models** $(\mathbb{C}, \dots)$ and $x \in \mathbb{C}$, $x \Vdash \phi$
then ϕ is provable in intuitionistic predicate logic

(Completeness requires a classical meta-theory.)

Coverage base

a.u.a. base for a
Grothendieck topology

on a category

A coverage base on a category $\mathcal{C}$ is a relation of the form $C \triangleright X$,

where $X \in \mathcal{C}$ and $C \subseteq \bigcup_{Y \in \mathcal{C}} \mathcal{C}(Y, X)$

Reflexivity

$$\{x \xrightarrow{\text{id}_x} x\} \triangleright x$$

Transitivity

If $C \triangleright X$ and, for all $Y \xrightarrow{f} X \in C$, $C_f \triangleright Y$

then $\{z \xrightarrow{g \circ f} X \mid Y \xrightarrow{f} X \in C, z \xrightarrow{g} Y \in C_f\} \triangleright X$

Stability

If $C \triangleright X$ and $X' \xrightarrow{f'} X$ then there exists $D \triangleright X'$ such that,
for all $Y' \xrightarrow{g'} X' \in D$ there exist $Y \xrightarrow{g} X \in C$ and $Y' \xrightarrow{f'} X'$

such that

$$\begin{array}{ccc} Y' & \xrightarrow{f'} & Y \\ g' \downarrow & & \downarrow g \\ X' & \xrightarrow{f} & X \end{array}$$

A separated model is given by

- A small category $\mathbb{C}$ of worlds with coverage base $\triangleright$
- To every $x \in \mathbb{C}$ a set A_x
- Transition functions $(\mapsto \cdot f : A_x \rightarrow A_y)_{y \xrightarrow{f} x}$ s.t. $(a \cdot f) \cdot g = a \cdot (f \circ g)$
- $\mathbb{C} \triangleright X \Rightarrow \forall a, b \in A_x (\forall y \xrightarrow{f} x \in \mathbb{C}, a \cdot f = b \cdot f) \Rightarrow a = b$. ($\triangleleft$ -separated)
- For every predicate P of arity k subsets $P_w \subseteq (A_w)^k$ satisfying
 - $\forall \vec{a} \in A_x^k (\vec{a} \in P_x \Rightarrow \forall y \xrightarrow{f} x, \vec{a} \cdot f \in P_y)$
 - $\mathbb{C} \triangleright X \Rightarrow \forall \vec{a} \in A_x^k (\forall y \xrightarrow{f} x \in \mathbb{C}, \vec{a} \cdot f \in P_y) \Rightarrow \vec{a} \in P_x$ (P is $\triangleleft$ -closed)

Category-theoretic Formulation

A separated Model is given by

- A small category $\mathcal{C}$ of Worlds with coverage base $\triangleright$
- A $\triangleright$ -separated presheaf $A : \mathcal{C}^{op} \rightarrow \underline{\text{Set}}$
- For every predicate P of arity k , a $\triangleright$ -closed subpresheaf $P \subseteq A^k$

Forcing w.r.t. a coverage

$$X \Vdash \phi \vee \psi \Leftrightarrow$$

there exists a cover $C \triangleright X$ such that,

for every $Y \xrightarrow{f} X \in C$, $Y \Vdash \phi \cdot f$ or $Y \Vdash \psi \cdot f$

$$X \Vdash \perp \Leftrightarrow \emptyset \triangleright X$$

$X \Vdash \exists x \phi \Leftrightarrow$ there exists a cover $C \triangleright X$ such that,

for every $Y \xrightarrow{f} X \in C$

there exists $a \in \underline{A}_Y$ s.t. $Y \Vdash (\phi \cdot f)[x := a]$

Meta theorems for sheaf semantics

Monotonicity If $x \Vdash \phi$ and $y \xrightarrow{f} x$ then $y \Vdash \phi \cdot f$

Cover property If $C \triangleright x$ satisfies, for all $y \xrightarrow{f} x \in C$, $y \Vdash \phi \cdot f$ then $x \Vdash \phi$.

Soundness If ϕ is provable in intuitionistic predicate logic then for all *separated models* $(C, \triangleright, \dots)$ and $x \in C$, $x \Vdash \phi$.

Completeness If, for all *separated models* $(C, \triangleright, \dots)$ and $x \in C$, $x \Vdash \phi$ then ϕ is provable in intuitionistic predicate logic

Completeness has an elegant proof in a constructive meta-theory

To prove completeness we use a category based model

The category $\mathbb{C}_{ctx}$

Objects: $\Theta; \Gamma$ (left-hand sides of sequents)

Morphisms: $\sigma: \Theta'; \Gamma' \rightarrow \Theta; \Gamma$

$\sigma: \Theta \rightarrow \Theta'$ is an injective function

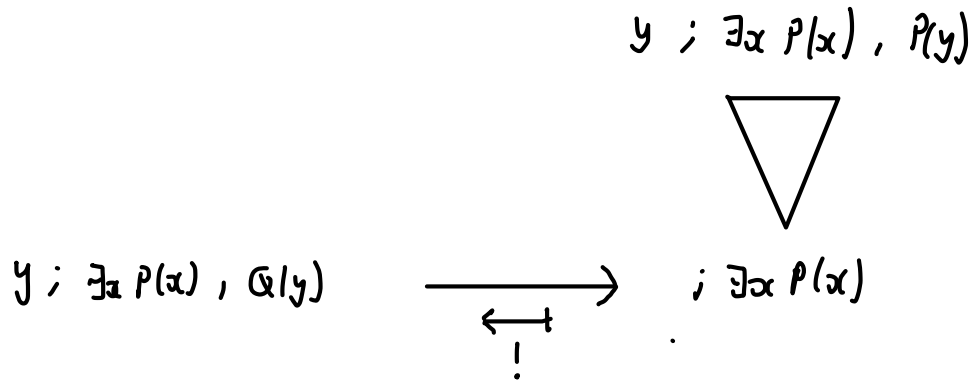
and $\Gamma[\sigma] \subseteq \Gamma'$

The coverage base $\triangleright_{ctx}$

$C \triangleright \Theta; \Gamma \iff \frac{C}{\Theta; \Gamma}$ is a derivable context rule.

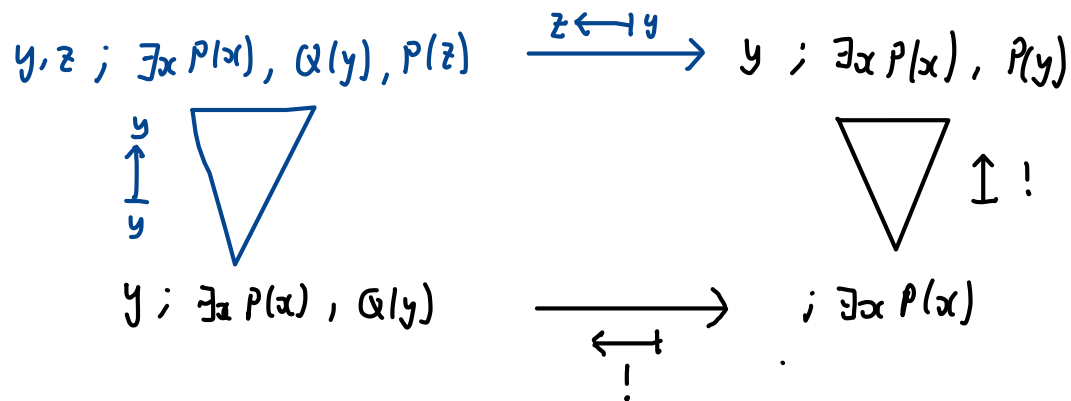
$\triangleright_{ctx}$ is a coverage base

Example illustrating stability



Δ_{ctx} is a coverage base

Example illustrating stability



Definition of full separated model $\mathcal{M}_{ctx}$

- Category of worlds $\mathcal{C}_{ctx}$
- Coverage base Δ_{ctx}
- Separated presheaf

$$A_{\theta; \Gamma} := \{t \mid \text{vars}(t) \subseteq \theta\} / \sim_{\theta; \Gamma}$$

$$t \sim_{\theta; \Gamma} t' :\Leftrightarrow \forall \theta'; \Gamma' \xrightarrow{\sigma} \theta; \Gamma \text{ and } \theta' \vdash [z] \text{-formula } \Psi[z]$$

$$\theta'; \Gamma' \vdash_{cf} \Psi[t[\sigma]] \Leftrightarrow \theta'; \Gamma' \vdash_{cf} \Psi[t'[\sigma]]$$

$$[t] \cdot \sigma := [t[\sigma]]$$

- Interpretation of predicates

$$([t_1], \dots, [t_n]) \in P_{\theta; \Gamma} :\Leftrightarrow \theta; \Gamma \vdash_{cf} P(t_1, \dots, t_n)$$

The proof of cut-free completeness

entailment w.r.t.

category-based separated models.

$$\Theta; \Gamma \models \psi \Rightarrow \Theta; \Gamma \vdash_{cf} \psi$$

now proceeds along similar lines to the propositional case

Once again, the main step is to prove

Main lemma $\mathcal{N}_{ctx}$ enjoys the following.

For any $\Theta; \Gamma \in \mathbb{C}_{ctx}$ and Θ -formula ϕ

(1) if $\phi \in \Gamma$ then $\Theta; \Gamma \Vdash \phi[\vec{x} := \overrightarrow{[x]}]$

(2) if $\Theta; \Gamma \Vdash \phi[\vec{x} := \overrightarrow{[x]}]$ then $\Theta; \Gamma \vdash_{cf} \phi$.

We have obtained completeness for intuitionistic predicate logic with respect to category-based separated models.

$$(\forall \mathcal{N} = (\mathcal{C}, \Delta, \dots), \theta; \Gamma \models_{\mathcal{N}} \psi) \Rightarrow \theta; \Gamma \vdash \psi$$

This can be strengthened to completeness with respect to poset-based separated models

$$(\forall \mathcal{M} = (\mathcal{W}, \Delta, \dots), \theta; \Gamma \models_{\mathcal{M}} \psi) \Rightarrow \theta; \Gamma \vdash \psi$$

using the **Diaconescu cover**, which can be used to construct a poset-based model out of a category-based model, preserving logical properties.

Sheaf models

Separated Model

- A small category $\mathcal{C}$ of worlds with coverage base $\triangleright$
- A Separated presheaf $A : \mathcal{C}^{op} \rightarrow \underline{\text{Set}}$
- For every predicate P of arity k , a closed subpresheaf $P \subseteq A^k$

Sheaf model (a special case of separated model)

- A small category $\mathcal{C}$ of worlds with coverage base $\triangleright$
- A sheaf $A : \mathcal{C}^{op} \rightarrow \underline{\text{Set}}$ (every sheaf is a fortiori separated)
- For every predicate P of arity k , a closed subpresheaf $P \subseteq A^k$ $\equiv$ subsheaf

Sheaf on a topological space T

Given a presheaf $A: \mathcal{O}(T)^{op} \rightarrow \underline{\text{Set}}$

- A matching family for a cover $C \triangleright w$ is a family $(a_v \in A_v)_{v \in C}$ such that $\forall u, v \in C \quad a_u \cdot (u \cap v) = a_v \cdot (u \cap v)$
- An amalgamation for a matching family $(a_v \in A_v)_{v \in C}$ is an element $a_w \in A_w$ s.t. $\forall v \in C \quad a_v = a_w \cdot v$.
- The presheaf A is a sheaf if every matching family has a unique amalgamation. (Separated $\Leftrightarrow$ at most one amalgamation!)

Example sheaf $C: \mathcal{O}(T) \rightarrow \underline{\text{Set}} \quad C(u) := \{f: u \rightarrow \mathbb{R} \mid f \text{ continuous}\}$
 $f \cdot v := f|_v \quad \text{for } f \in C(u) \text{ and } v \subseteq u$

Sheaf for a coverage base (over a poset)

Given a presheaf $A: W^{op} \rightarrow \underline{\text{Set}}$ and coverage base $\triangleright$ on W .

- A matching family for a cover $C \triangleright w$ is a family $(a_v \in A_v)_{v \in C}$ such that $\forall v, v' \in C \quad \forall u \leq v, v', \quad a_v \cdot u = a_{v'} \cdot u$
- An amalgamation for a matching family $(a_v \in A_v)_{v \in C}$ is an element $a_w \in A_w$ s.t. $\forall v \in C \quad a_v = a_w \cdot v$.
- The presheaf A is a sheaf if every matching family has a unique amalgamation. (Separated $\Leftrightarrow$ at most one amalgamation!)

Sheaf for a coverage base on a category

Given a presheaf $A: \mathcal{C}^{op} \rightarrow \underline{\text{Set}}$ and coverage $\triangleright$ on $\mathcal{C}$.

- A matching family for a cover $C \triangleright X$ is a family $(a_f \in A_Y)_{Y \xrightarrow{f} X \in C}$ such that $\forall Y \xrightarrow{f} X, Y' \xrightarrow{f'} X \in C, \forall Z \xrightarrow{g} Y$ in $\mathcal{C}$, $a_f \cdot g = a_{f'} \cdot g'$.

$$\begin{array}{ccc} & g & \\ & \downarrow & \\ g' & \downarrow & \\ Y' & \xrightarrow{f'} & X \end{array}$$
- An amalgamation for a matching family $(a_f \in A_Y)_{Y \xrightarrow{f} X \in C}$ is an element $a \in A_X$ s.t. $\forall Y \xrightarrow{f} X \in C, a_f = a \cdot f$.
- The presheaf A is a sheaf if every matching family has a unique amalgamation. (Separated $\Leftrightarrow$ at most one amalgamation!)

There is no loss of generality in considering sheaf models

Theorem

Let $(\mathbb{C}, \triangleright)$ be a site (small category + coverage (base)).

Every separated model $\mathcal{N}$ over $(\mathbb{C}, \triangleright)$ has a dense elementary embedding into a sheaf model $\mathcal{N}^+$ over $(\mathbb{C}, \triangleright)$.

$$\mathcal{N} \hookrightarrow \mathcal{N}^+$$

Moreover, this characterises $\mathcal{N}^+$ up to isomorphism.

Given presheaves $A, B : \mathbb{C}^{op} \rightarrow \underline{\text{Set}}$

A presheaf map $\alpha : A \rightarrow B$ is a family of functions

$$(\alpha_x : A_x \rightarrow B_x)_{x \in \mathbb{C}}$$

such that $\forall y \xrightarrow{f} x$ in $\mathbb{C}$ and $a \in A_x$, $\alpha_y(a \cdot f) = (\alpha_x(a)) \cdot f$.

I.e., a presheaf map is a *natural transformation* $\alpha : A \Rightarrow B$.

α is an embedding of presheaves if every α_x is injective.

α is dense w.r.t. $\triangleright$ if

$$\forall b \in B_x \exists C \triangleright x \text{ s.t. } \forall Y \xrightarrow{f} X \in \mathbb{C}, \exists a \in A_Y \text{ s.t. } \alpha_Y(b) = a \cdot f.$$

Given separated models $\mathcal{N} = (A, (P)_P)$ and $\mathcal{N}' = (A', (P')_P)$

an embedding of models is a presheaf embedding $\alpha: A \rightarrow A'$ that further satisfies

For all $x \in \mathbb{C}$ and $a_1, \dots, a_n \in A_x$

$$(a_1, \dots, a_n) \in P_x \iff (\alpha_x(a_1), \dots, \alpha_x(a_n)) \in P'_x$$

It is an elementary embedding if, for all formulas ϕ ,

$$x \Vdash^{\mathcal{N}} \phi(a_1, \dots, a_n) \iff x \Vdash^{\mathcal{N}'} \phi(\alpha_x(a_1), \dots, \alpha_x(a_n))$$

Grothendieck's $(\cdot)^+$ construction

Let $\mathbb{C}$ be a small category with coverage base $\triangleright$.

Given a presheaf $A: \mathbb{C} \rightarrow \underline{\text{Set}}$ define $A^+ : \mathbb{C} \rightarrow \underline{\text{Set}}$

$$A_x^+ := \left\{ (c, (a_f)_{f \in c}) \mid c \triangleright x, (a_f) \text{ is a matching family for } c \right\} / \sim$$

$$(c, (a_f)_f) \sim (c', (a'_f)_{f'}) \iff \text{for every commuting square in } \mathbb{C}$$

$$\begin{array}{ccc} \bullet & \xrightarrow{g} & \bullet \\ g' \downarrow & & \downarrow f \in c \\ \bullet & \xrightarrow{f' \in c'} & x \end{array} \quad , \quad a_f \cdot g = a'_{f'} \cdot g'$$

Theorem (Grothendieck)

- A^+ is separated, for any presheaf A .
- If A is separated, then A^+ is a sheaf,
and $a \in A_x \mapsto [(\{\text{id}_x\}, \{a\})] : A \rightarrow A^+$
is a dense embedding of presheaves.

Extending a model

Theorem

Suppose $\alpha: A \Rightarrow A'$ is a dense embedding between separated presheaves.

Let $\mathcal{N} = (A, (P)_P)$ be any separated model.

For any k -ary predicate P , define $P' \subseteq (A')^k$ by

$$(a'_1, \dots, a'_k) \in P'_x : \Leftrightarrow \exists C \supset X \text{ s.t. } \forall y \xrightarrow{f} x \in C$$

$$\exists a_1, \dots, a_k \in A_y . \alpha_y(a_1) = a'_1 \cdot f , \dots , \alpha_y(a_k) = a'_k \cdot f \\ \text{and } (a_1, \dots, a_k) \in P_y$$

Then $\mathcal{N}' := (A', (P')_P)$ is a separated model and $\alpha: \mathcal{N} \Rightarrow \mathcal{N}'$ is an elementary embedding.

Proof

We just look at the elementary embedding property.

We prove

$$X \models^N \phi(a_1, \dots, a_n) \iff X \models^{N'} \phi(\alpha_X(a_1), \dots, \alpha_X(a_n))$$

by induction on ϕ .

We consider just one case.

Case $\phi \equiv \exists x \phi'$

$\Rightarrow$ Suppose $X \Vdash (\exists x \phi')[a_1, \dots, a_n]$

There exists $C \triangleright X$ s.t. $\forall Y \xrightarrow{F} X \in C, \exists a \in X_Y$ s.t.

$$Y \Vdash^N \phi'[a_1 \cdot f, \dots, a_n \cdot f, a]$$

By i.h., $Y \Vdash^{N'} \phi'[\alpha_Y(a_1 \cdot f), \dots, \alpha_Y(a_n \cdot f), \alpha_Y(a)]$

i.e., $Y \Vdash^{N'} \phi'[\alpha_X(a_1) \cdot f, \dots, \alpha_X(a_n) \cdot f, \alpha_Y(a)]$

Hence $X \Vdash^{N'} (\exists x \phi')[\alpha_X(a_1), \dots, \alpha_X(a_n)]$.

$\Leftarrow$

Suppose $X \Vdash^{N'} (\exists x \phi') [\alpha_x(a_1), \dots, \alpha_x(a_n)]$

There exists $C \triangleright X$ s.t. $\forall Y_f \xrightarrow{f} X, \exists a'_f \in A'_{Y_f}$ s.t.

$$Y_f \Vdash^{N'} \phi' [\alpha_x(a_1) \cdot f, \dots, \alpha_x(a_n) \cdot f, a'_f]$$

By density $\exists C_f \triangleright Y_f \forall z \xrightarrow{g} Y_f \in C_f \exists a_g \in A_z \alpha_z(a_g) = a'_f \cdot g$.

By transitivity of $\triangleright$, $D := \{f \circ g \mid f \in C, g \in C_f\} \triangleright X$ warning!

Moreover, $\forall z \xrightarrow{f \circ g} X \in D, z \Vdash^{N'} \phi' [\alpha_z(a_1 \cdot f \cdot g), \dots, \alpha_z(a_n \cdot f \cdot g), \alpha_z(a_g)]$.

By i.h., $z \Vdash^N \phi' [a_1 \cdot f \cdot g, \dots, a_n \cdot f \cdot g, a_g]$

Hence, $z \Vdash^N (\exists x \phi') [a_1, \dots, a_n]$.

□

The **warning** is that $\{f \circ g \mid f \in C, g \in C_f\}$ is not a good definition, as it uses choice to select a chosen C_f for every f .

(Similar issues arise in proving that Π enjoys the cover property in the case of $\forall$ and $\exists$.)

Two possible solutions

- (recommended) Define Π w.r.t. a coverage rather than coverage base.
- (this might work?!) Formulate coverage bases using indexed families

$$\{Y_i \xrightarrow{f_i} X\}_{i \in I} \triangleright X.$$

Perspective

Coverage $\equiv$ Grothendieck topology

Coverage base $\equiv$ base for a Groth. top. (one sees other variant definitions)

The category of presheaf maps between sheaves is a Groth. topos.

All Groth. toposes arise in this way.

A sheaf model over $(\mathbb{C}, \triangleright)$ is just

- An object $A \in \mathcal{E} := \underline{\text{Sh}}(\mathbb{C}, \triangleright)$
- For every predicate P a subobject $P \twoheadrightarrow A^k$ in $\mathcal{E}$

That is, it is an ordinary model from the perspective of $\mathcal{E}$.

The forcing relation captures truth in the internal logic of $\mathcal{E}$.

E.g.

$$\forall x \in \mathbb{C}, x \Vdash \phi \quad \Leftrightarrow \quad \mathcal{E} \text{ validates } \mathcal{V} \models \phi$$

Internal
Tarskian
semantics

Literature

The literature I know on sheaf semantics approaches it via first understanding toposes and their logic, requiring substantial category theory.

My favourite book that takes this approach is

- Mac Lane & Moerdijk *Sheaves in Geometry and Logic*

A more gently paced presentation (with less content) can be found in

- Goldblatt *Topoi*